India Ocean Freight Index

IOFI Monthly Report & 3-Month Lane Forecast — v2 (Data-Driven Model)

July 2026 | Nhava Sheva, ICD Tihi, ICD Dhannad → 60 destination ports | 180 lanes

Current IOFI: **162.7** 3-mo model median: **168.4** (+3.5%) 5th–95th pct range: 155.9 – 181.8

Executive Summary

The IOFI stands at 162.7 this month. This release replaces the fixed-weight, hand-decayed version of the model with a fully data-driven pipeline: driver weights are learned (Ridge, ElasticNet, Bayesian Ridge, XGBoost, ensembled and cross-checked with SHAP), each macro driver is forecast independently with an auto-selected time-series model (ARIMA / ETS / Kalman filter), lanes are grouped by an automatic clustering step, and uncertainty is expressed as statistically estimated percentile bands rather than hand-set High/Base/Low scenarios.

The single largest learned contributor to the current index level is **war risk premium idx**. Under the model's median forecast path, the index reaches **168.4** (+3.5%) within three months, with a statistically estimated 5th–95th percentile range of 155.9 to 181.8. Unlike the prior version, this path is not assumed to decay — it is the output of the driver-level forecasts feeding back through the learned index model, so it can just as easily continue rising as it can fall, depending on what the underlying time-series models project for each driver.

Across the 180 India-origin lanes, the model-median case implies moves ranging from -10.0% to +-3.9%. Houston (Nhava Sheva (JNPT)) shows the largest projected decline (-10.0%), while Karachi (Nhava Sheva (JNPT)) shows the largest projected increase (+-3.9%). In backtesting over 8 walk-forward months, the learned model achieved a 2.9% MAPE (bootstrap 95% CI: 1.3%–4.9%) and 88% directional accuracy, versus 6.3% MAPE for naive persistence. With this few backtest folds the CI is itself wide -- treat the point MAPE as directional, not precise.

1. IOFI: History & Statistically Learned 3-Month Forecast

The IOFI is a composite of 12 macro/structural drivers of India-origin container freight rates -- ten carried over from v2.0 plus two China-side supply drivers added in v2.1 (Section 3). Base = 100 at the start of the history window (Aug 2024). Instead of a hand-picked scenario decay, each driver is forecast independently (ARIMA/ETS/Kalman filter, auto-selected by AIC), the forecasted driver paths are fed through the learned index model (Ridge/ElasticNet/Bayesian Ridge/XGBoost ensemble) to get a median path, and a bootstrap ensemble produces 5th/25th/50th/75th/95th percentile bands.

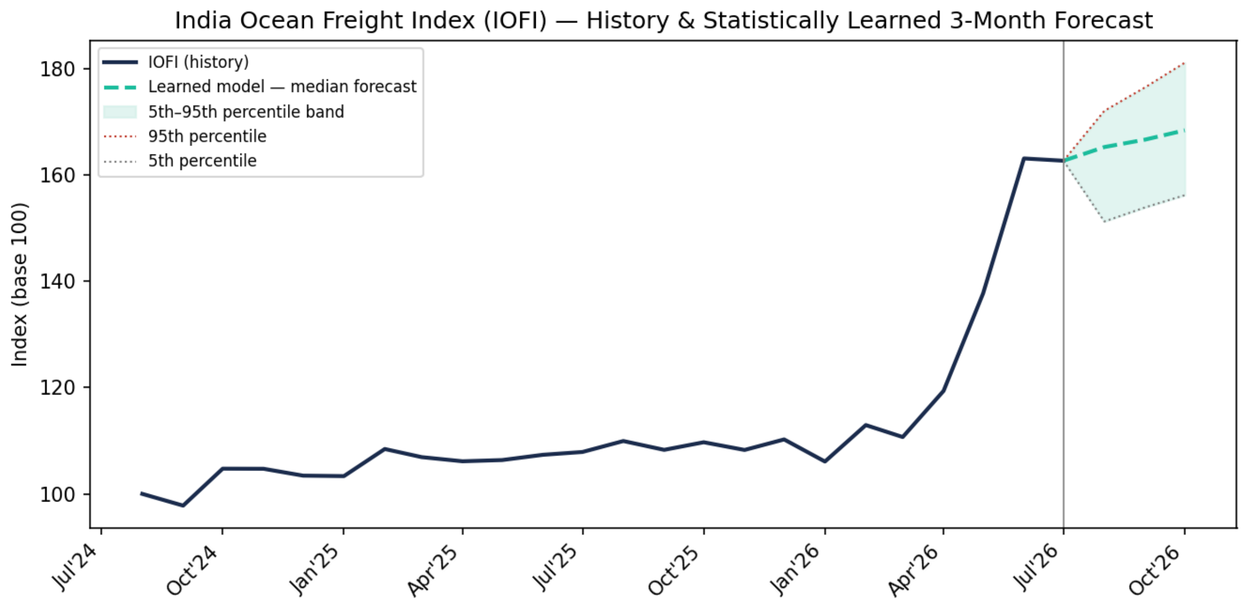

Figure 1. IOFI history and 3-month statistically forecasted percentile band.

Month	P5	P25	Median (P50)	P75	P95
2026-08	152.0	160.8	166.6	169.3	172.7
2026-09	154.2	164.1	170.6	173.6	177.3
2026-10	155.9	166.7	174.0	177.4	181.8

2. Driver Attribution — Learned, Not Assigned

Each driver's contribution is estimated, not hand-weighted. Ridge, ElasticNet, Bayesian Ridge and XGBoost are each fit on the engineered feature set (lags, rolling stats, momentum, seasonality, nonlinear interactions) and their normalized importances are ensembled. XGBoost's contribution is cross-checked with SHAP values so the ranking below reflects genuine learned sensitivity rather than an analyst's prior.

Sample-size guard on XGBoost: at n=22 training months, XGBoost's vote in the ensemble average is scaled by 0.282x rather than counted at full weight (full weight requires n≥60 months) -- boosted trees are the estimator most prone to overfitting at this sample size, so it is discounted rather than dropped: it remains the only estimator here that captures nonlinear driver interactions without those interactions being hand-specified. Raw, unshrunk XGBoost importances are retained in learned_driver_weights.csv for full transparency.

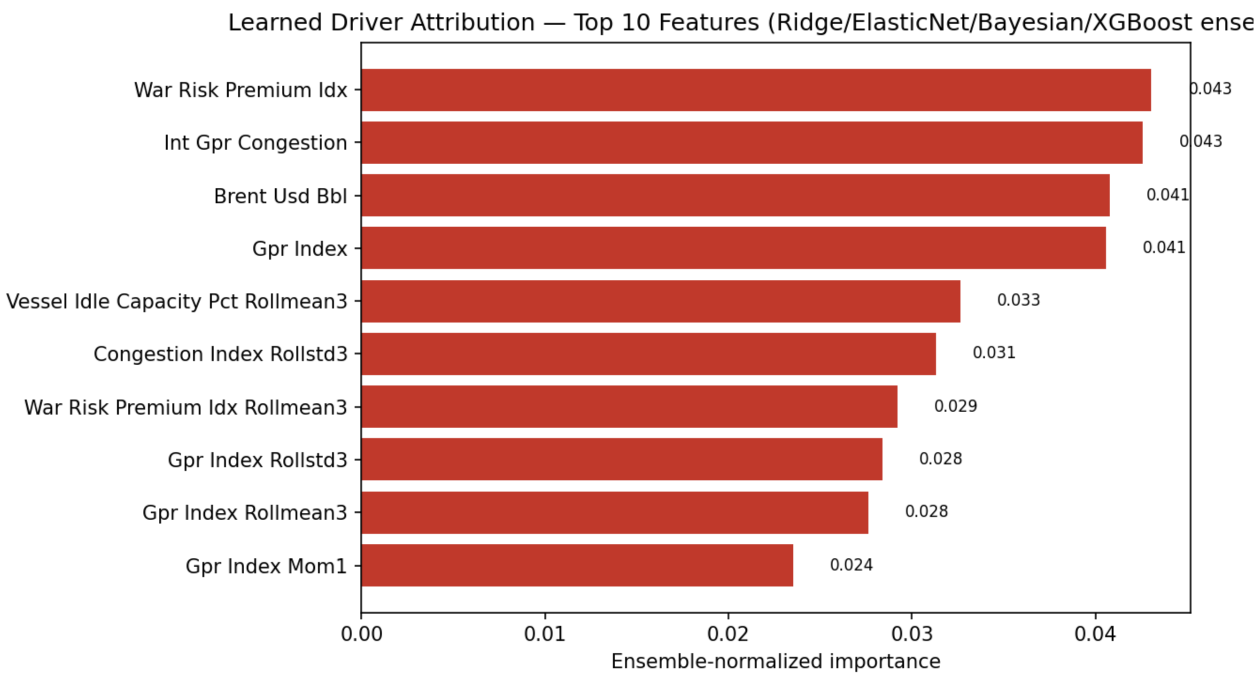

Figure 2. Top 10 learned features by ensemble-normalized importance (sample-size-adjusted).

3. Driver Deep-Dives

v2.1 adds two China-side supply drivers to the 10 carried in v2.0 -- a manufacturing-PMI-like index and an export container throughput index -- so the model captures China's own festival calendar (Chinese New Year factory shutdowns, Golden Week) and export capacity directly, rather than only inferring China-side effects indirectly through congestion and idle capacity. See Figure 6b.

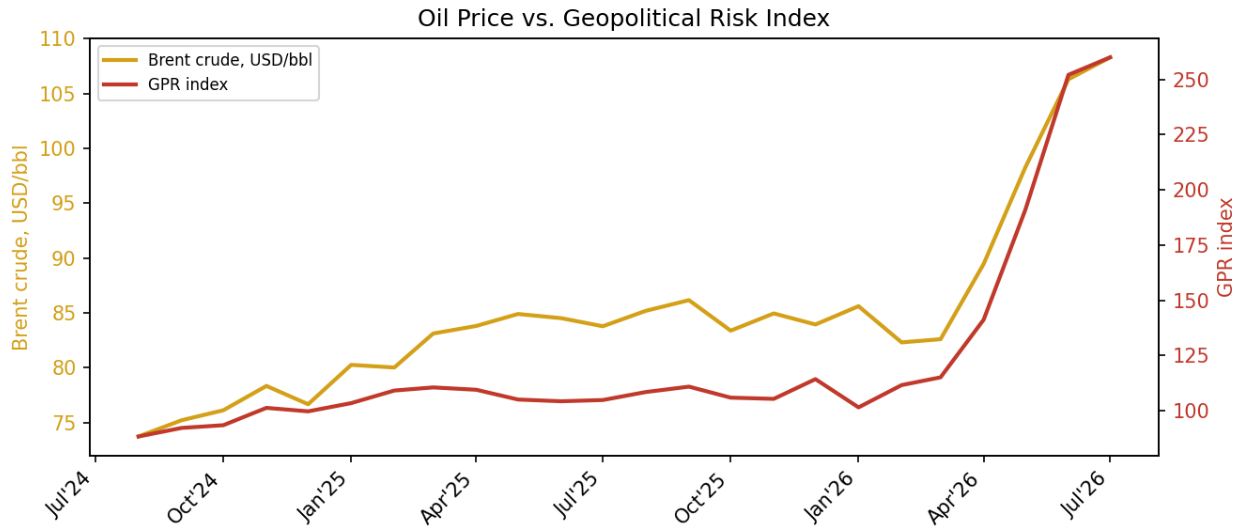

Figure 3. Oil price vs. geopolitical risk index — both learned as top contributors.

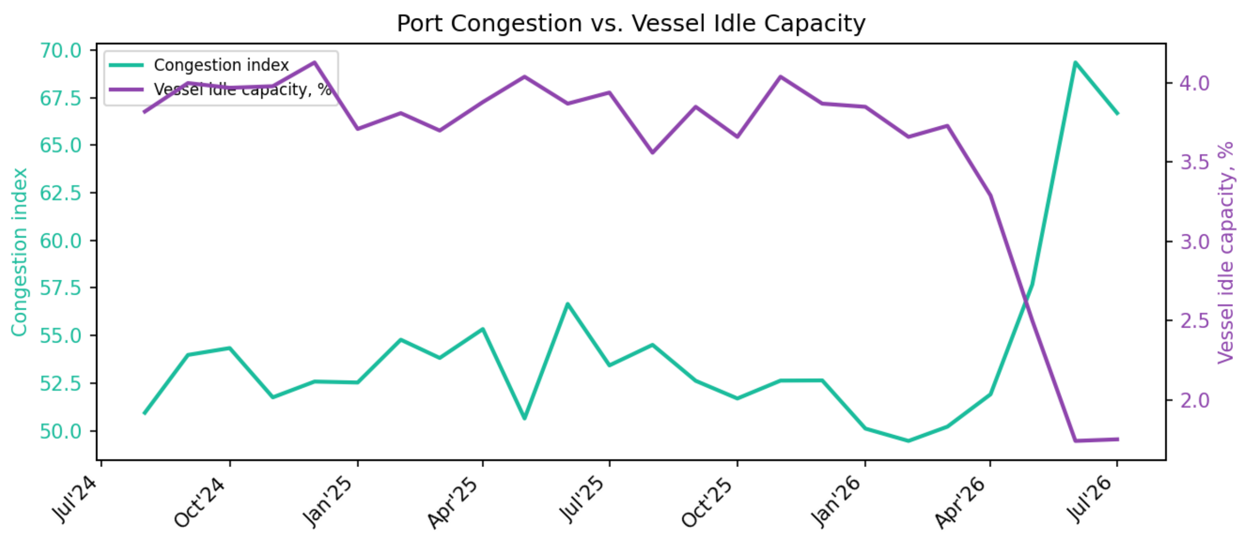

Figure 4. Port congestion vs. vessel idle capacity (idle capacity carries a learned inverse relationship).

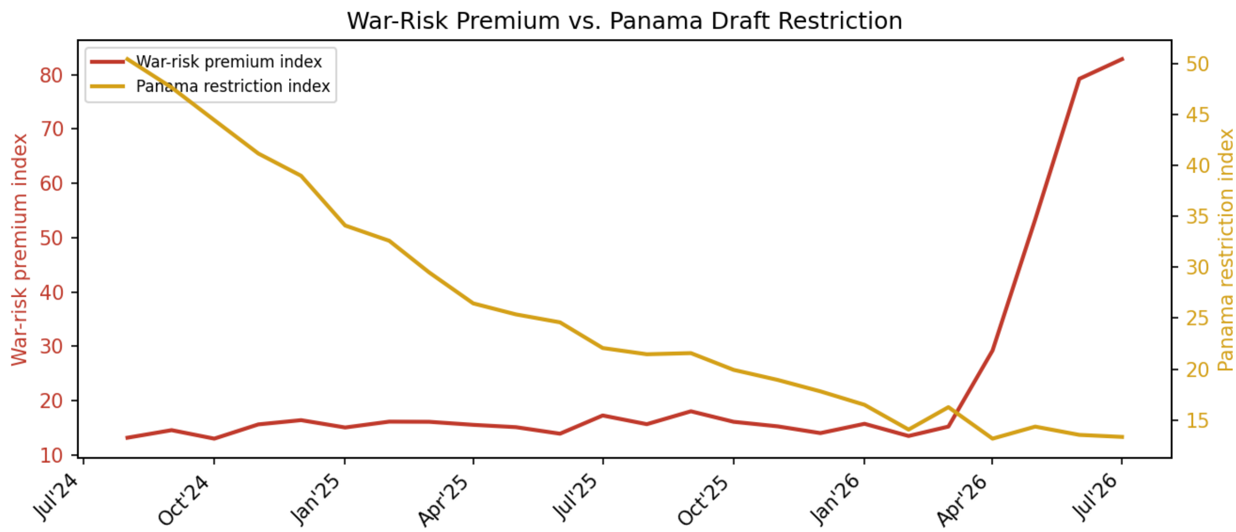

Figure 5. War-risk insurance premium vs. Panama Canal draft restriction — two structurally distinct drivers.

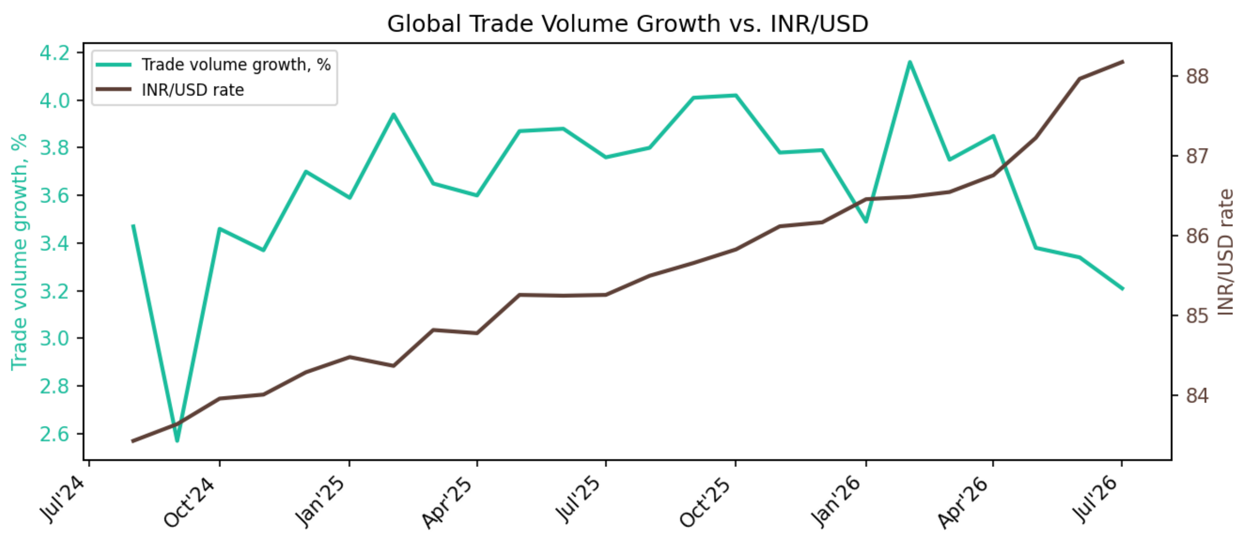

Figure 6. Global trade volume growth vs. INR/USD.

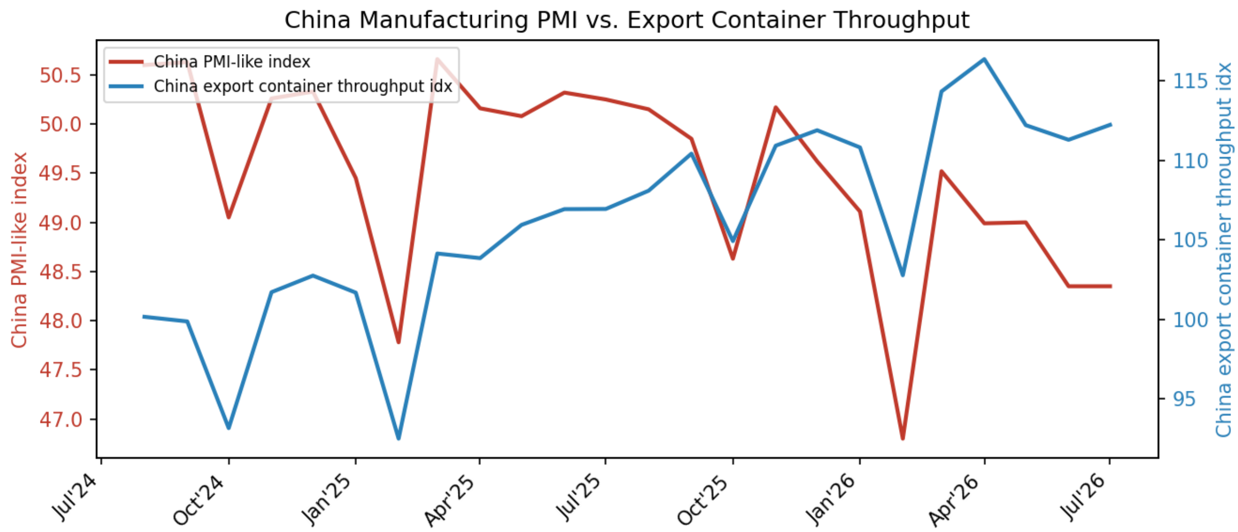

Figure 6b. China manufacturing PMI-like index vs. export container throughput — new in v2.1. Both dip sharply every February (Chinese New Year factory shutdowns) and, to a lesser extent, every October (Golden Week); throughput growth stalls through the recent congestion spike as capacity gets absorbed rather than cleared.

4. Lane-Level Rate Forecasts — Hierarchical Global → Regional → Seasonal → Lane

Lane translation is now structured in three learned levels plus one calendar-driven adjustment, instead of one flat step. The global IOFI move feeds a **regional** move (the median historical pass-through of all lanes within a destination region — 10 regions in total), and each lane's own historical pass-through is expressed as a **lane premium** on top of its region: the amount that lane moves beyond what its corridor alone would explain. On top of both, a **regional seasonal premium** adds the festival/holiday calendar effect relevant to that specific destination region and forecast month — Diwali for Indian Ocean Rim, Golden Week and Chinese New Year for Intra-Asia (the China-side counterpart to the existing US import-season/Black Friday/Christmas seasonality), Ramadan/Eid for Middle East and Red Sea/East Africa, and US import-season peak for the US-bound regions — capped at ±2.5% per month so it nudges the learned forecast rather than overriding it. The regional, lane-premium, and seasonal components sum back to the total forecast (see lane_forecasts_v2.csv columns regional_pct_change / lane_premium_pct_change / seasonal_pct_change), letting a planner see whether a lane's move is “this corridor is repricing,” “this specific lane is unusual for its corridor,” or “this month's festival calendar for this region.” For reference, lanes are also grouped into 7 automatically discovered KMeans clusters (silhouette score 0.735) using distance/route-exposure/chokepoint/volatility features.

Region	# Lanes	Region Beta	Min Premium	Median Premium	Max Premium
Middle East	18	2.61	-0.04	+0.00	+0.06
Red Sea / East Africa	15	2.59	-0.06	+0.00	+0.06
Europe (North)	15	2.45	-0.05	+0.00	+0.14
Europe (Mediterranean)	21	2.45	-0.10	+0.00	+0.14
US East Coast	15	1.52	-0.13	+0.00	+0.11
US Gulf / Caribbean / Latin America	15	1.36	-0.10	+0.00	+0.10
Indian Ocean Rim	15	0.90	-0.08	+0.00	+0.12
Africa / South America (Cape)	18	0.72	-0.13	+0.00	+0.11
Pacific (US West / Oceania)	21	0.72	-0.11	+0.00	+0.23
Intra-Asia	27	0.55	-0.19	+0.00	+0.16

Table: regional pass-through beta (how much the whole corridor co-moves with the index) and the spread of individual lane premiums within each region (how much within-corridor heterogeneity a flat per-lane model would otherwise hide).

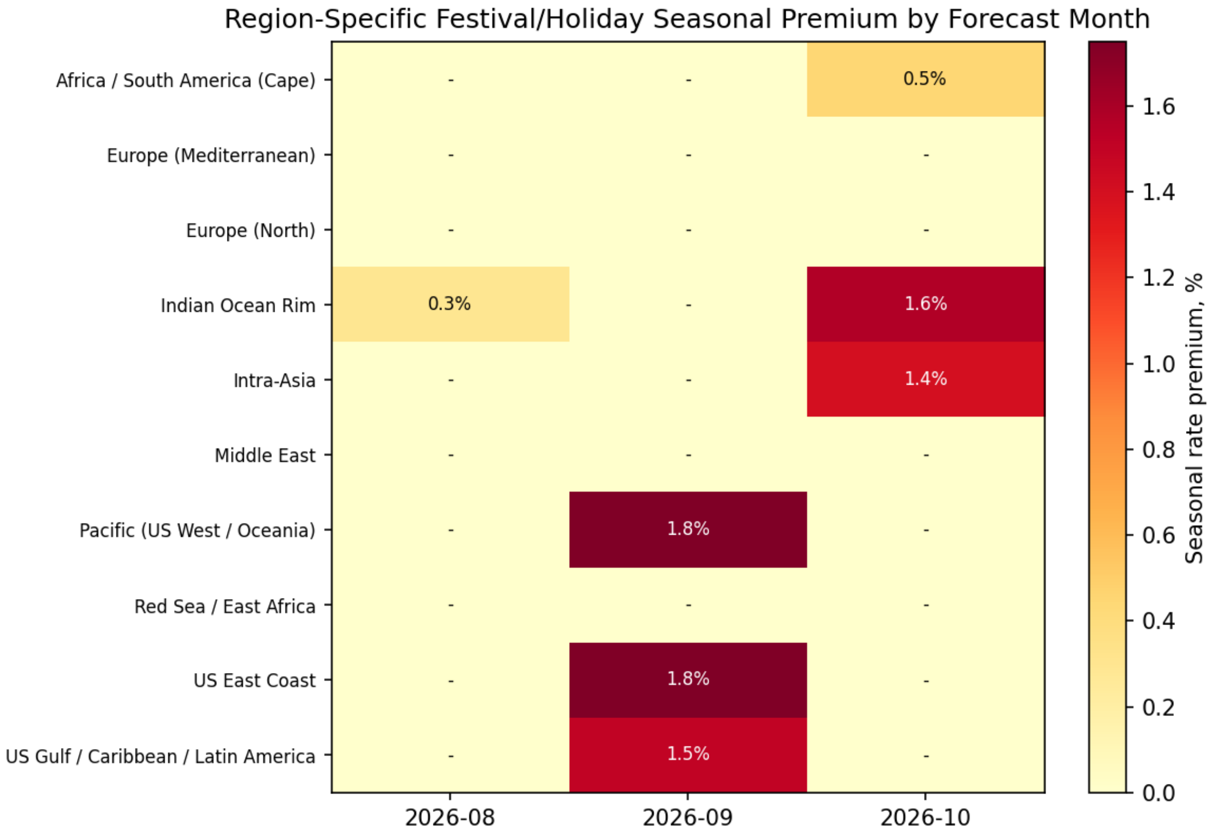

Figure 6c. Region-specific festival/holiday seasonal premium for each forecast month — the explicit “which festivals move which regions, and when” view. See seasonal_calendar_summary.csv for the named events behind each cell.

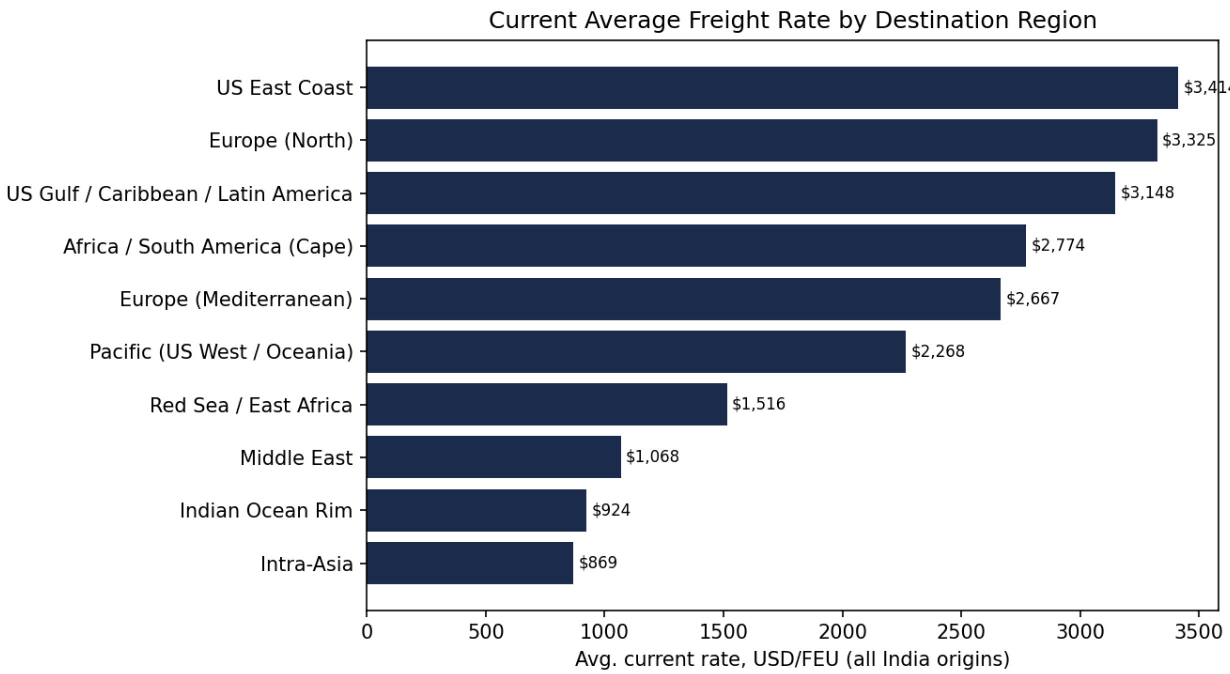

Figure 7. Current average freight rate by destination region (all India origins).

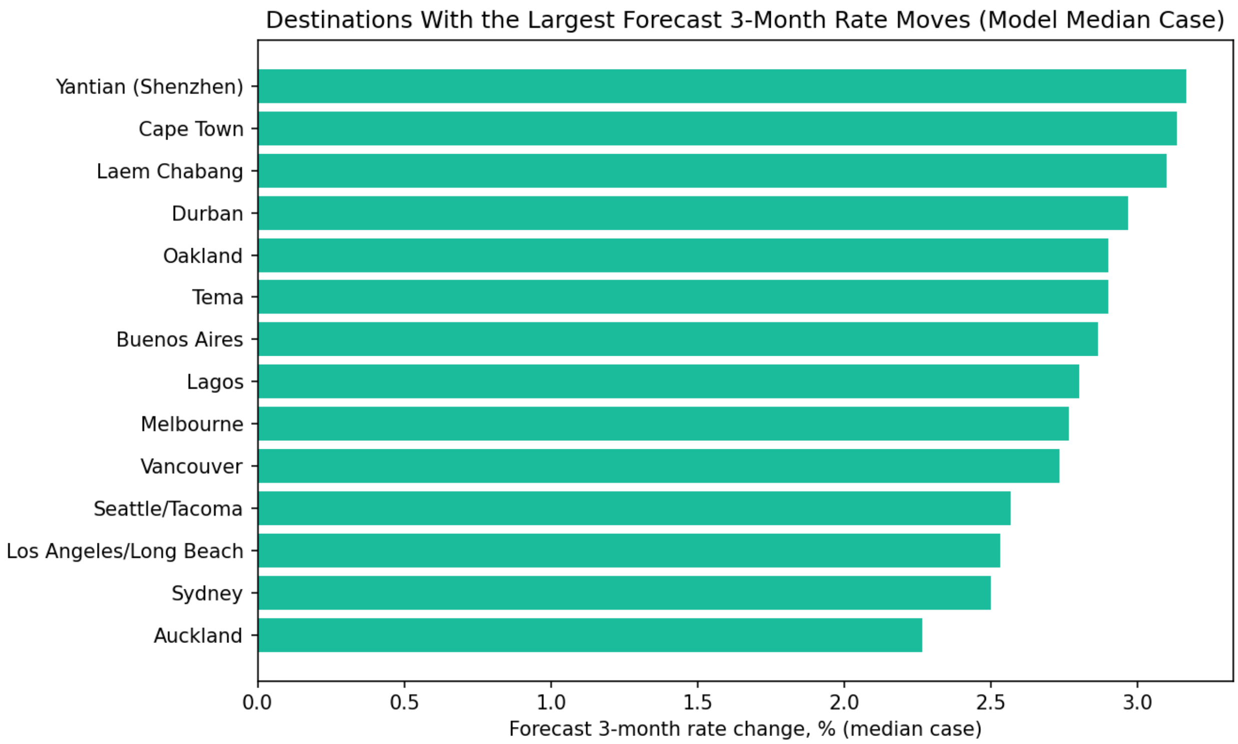

Figure 8. Destinations with the largest forecast 3-month rate moves, model median case.

5. Selected Lane Detail (Model Median Case, 3-Month Horizon)

Largest projected declines

Origin	Destination	Current (USD/FEU)	3-mo Forecast (USD/FEU)	% Change
Nhava Sheva (JNPT)	Houston	$7,909	$7,118	-10.0%
Nhava Sheva (JNPT)	Norfolk	$7,351	$6,616	-10.0%
Nhava Sheva (JNPT)	Savannah	$6,903	$6,213	-10.0%
Nhava Sheva (JNPT)	Baltimore	$7,920	$7,128	-10.0%
Nhava Sheva (JNPT)	Charleston	$7,254	$6,529	-10.0%
Nhava Sheva (JNPT)	New Orleans	$7,666	$6,899	-10.0%
Nhava Sheva (JNPT)	New York/New Jersey	$7,024	$6,322	-10.0%
ICD Tihi	Savannah	$7,319	$6,587	-10.0%
ICD Tihi	Baltimore	$8,336	$7,502	-10.0%
ICD Tihi	New York/New Jersey	$7,441	$6,697	-10.0%

Largest projected increases

Origin	Destination	Current (USD/FEU)	3-mo Forecast (USD/FEU)	% Change
Nhava Sheva (JNPT)	Karachi	$659	$633	+-3.9%
Nhava Sheva (JNPT)	Yantian (Shenzhen)	$811	$779	+-3.9%
Nhava Sheva (JNPT)	Ho Chi Minh City (Cat Lai)	$662	$636	+-3.9%
ICD Dhannad	Yokohama	$1,414	$1,357	+-4.0%
ICD Dhannad	Shanghai	$1,164	$1,117	+-4.0%
ICD Dhannad	Ho Chi Minh City (Cat Lai)	$1,031	$990	+-4.0%
ICD Dhannad	Laem Chabang	$996	$956	+-4.0%
ICD Dhannad	Port Klang	$860	$826	+-4.0%
ICD Dhannad	Singapore	$897	$861	+-4.0%
ICD Dhannad	Port Louis	$1,433	$1,376	+-4.0%

6. India Origins

Origin	State	Gateway Port	Inland Surcharge (USD/FEU)
Nhava Sheva (JNPT)	Maharashtra	Nhava Sheva	$0
ICD Tihi	Gujarat	Mundra	$185
ICD Dhannad	Madhya Pradesh	Mundra	$335

ICD Tihi and ICD Dhannad both gateway through Mundra; inland haulage surcharges reflect rail/trucking distance from each ICD to the port. Nhava Sheva (JNPT) is itself a gateway port, so no inland surcharge applies.

7. Backtesting — Rolling-Origin Validation

The learned model is validated with rolling-origin (walk-forward) backtesting: at each step, the model is trained only on data available up to that point and scored on the next unseen month. It is benchmarked against naive persistence and a 3-month moving average.

Fold detail: 22 total months in history; expanding window starts at 14 training months and walks forward through 8 test folds (one held-out month per fold). MAPE_CI95 below is a fold-resampling bootstrap 95% interval around each MAPE — with only 8 folds this interval is wide by construction; read it as a reminder of how much a single new data point could move the headline number, not as a tight bound.

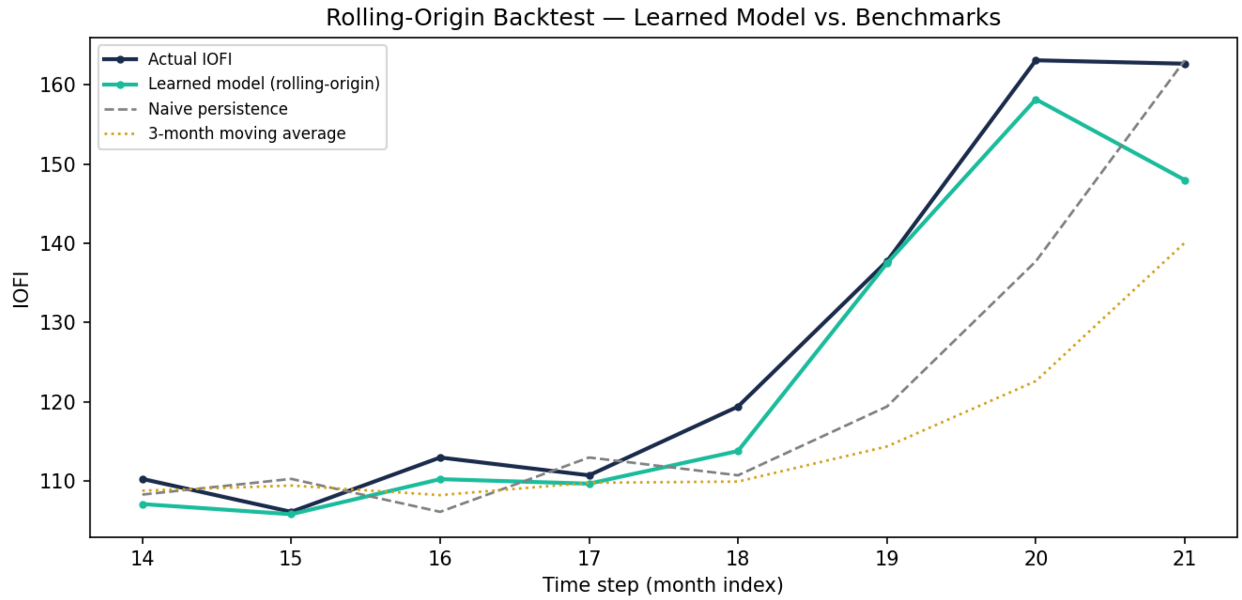

Figure 9. Rolling-origin backtest: learned model vs. naive persistence vs. moving average.

Model	MAPE	MAPE 95% CI	MAE	RMSE	Directional Accuracy
Learned model	2.93%	[1.3%, 4.9%]	4.09	6.02	88%
Naive persistence	6.28%	[3.0%, 10.1%]	8.51	11.88	0%
3-month moving avg	9.15%	[3.9%, 14.7%]	13.30	18.79	62%

Prediction interval calibration

Do the 5th–95th and 25th–75th percentile bands actually contain the realized value as often as their nominal coverage claims? This retrospectively rebuilds the same bootstrap-ensemble interval used for the live forecast at each backtest fold (trained only on data available at that point) and checks whether the true next-month IOFI fell inside it.

Interval	Nominal Coverage	Observed Coverage	Folds Inside / Total
P5-P95	90%	100.0%	8 / 8
P25-P75	50%	50.0%	4 / 8

Calibration estimated over 8 walk-forward folds -- small-sample, so read the observed coverage rate as directional (badly miscalibrated vs. roughly on-target), not as a precise percentage.

8. Methodology & Limitations

Index construction

The IOFI is a composite of 12 macro/structural drivers -- the original ten (oil, war-risk premium, GPR, congestion, vessel idle capacity, trade volume growth, USD index, INR/USD, Panama restriction, EU ETS carbon) plus two China-side supply drivers added in v2.1 (china_pmi_idx, china_export_container_idx) -- engineered into lags, rolling statistics, momentum, an expanded festival/holiday seasonality calendar, and nonlinear interaction terms (Oil×Congestion, WarRisk×Panama, TradeVolume×IdleCapacity, USD×Oil, GPR×Panama, and the new China PMI×Congestion / China Exports×IdleCapacity). Weights are learned — not assigned — via an ensemble of Ridge, ElasticNet, Bayesian Ridge and XGBoost, cross-checked with SHAP.

Festival/holiday calendar — expanded and region-aware

v2.0 carried a single global seasonality calendar anchored mostly to US demand timing (Black Friday, Christmas, US import season) plus two China-side supply events (Chinese New Year, Golden Week) as flat 0-1 monthly intensity features for the composite index. v2.1 keeps those as index-level features and adds Diwali, Ganesh Chaturthi, Eid al-Fitr, Eid al-Adha, Thanksgiving, and Singles Day (the China-side demand counterpart to Black Friday) to the calendar. It also adds a second, explicitly region-aware layer (REGION_SEASONAL_SENSITIVITY in feature_engineering.py) that maps each destination region to the festivals that actually move demand or supply for it -- Ramadan/Eid for Middle East and Red Sea/East Africa, Diwali for Indian Ocean Rim, Chinese New Year/Golden Week/Singles Day for Intra-Asia, Christmas/Black Friday/Thanksgiving for the US- and Europe-bound regions -- so a lane's forecast reflects the festival calendar relevant to where it's actually going, not a single blended seasonality applied uniformly across all 180 lanes. This region-aware layer feeds the capped seasonal premium described below in Lane-level translation.

Forecast methodology

Each driver is forecast independently using ARIMA, ETS (damped trend), or a Kalman-filter local linear trend model (the lightweight Bayesian Structural Time Series analogue used here), with the lowest-AIC model auto-selected per driver — no manual decay curve is assumed. The forecasted driver paths are passed through the learned index model to produce the composite IOFI median path, and a 400-iteration bootstrap ensemble produces the 5th–95th percentile bands.

Lane-level translation — hierarchical Global → Regional → Seasonal → Lane

Each lane's implied pass-through beta is calibrated directly from its own historical move relative to the index move that produced it (as in v2.0), then split into two layers: a **regional beta** (median lane beta within that destination region — the corridor-wide co-movement) and a **lane premium** (this lane's own beta minus its regional beta — how much it deviates from its corridor). The regional and lane-premium moves sum to the same total forecast as a flat per-lane beta, but the split is the structural piece requested for a future model: once monthly lane-level snapshots accumulate into a panel, the SAME regional grouping can be re-estimated as genuine fixed-effects or hierarchical-Bayes regional coefficients instead of a median-of-betas proxy — only the estimation method changes, not the Global → Regional → Lane structure itself. Lanes are additionally assigned to automatically discovered KMeans clusters for the driver-similarity view in Section 4, though — as in v2.0 — a genuine per-cluster regression sensitivity is not identifiable from a single cross-sectional snapshot (fitting Ridge per cluster on one time point returns near-zero R^2 because the macro drivers do not vary within it); this remains the primary open item on the roadmap.

v2.1 adds a fourth component, the **regional seasonal premium**: for each lane's destination region and forecast month, it sums (region's sensitivity to a festival) × (that festival's calendar intensity that month) across every active festival, then scales the result to a percentage capped at ±2.5% per month (hierarchical.py: regional_seasonal_pct). This is deliberately a transparent rule-based adjustment, not a learned one — with 22-24 months of history there isn't enough repetition of any single festival to fit its region-specific price impact directly, so the sensitivity weights (REGION_SEASONAL_SENSITIVITY in feature_engineering.py) are assigned by trade logic (Ramadan matters for Middle East demand, Golden Week matters for Intra-Asia capacity, etc.) rather than estimated. As monthly lane-level history accumulates, these can be replaced by region×festival coefficients fit the same way the regional beta itself will eventually be fit — same roadmap item as above.

XGBoost at small sample size

At n=22 training months, boosted trees are the estimator in this ensemble most prone to overfitting — linear/Bayesian shrinkage models degrade more gracefully at this sample size. Rather than remove XGBoost outright, its ensemble vote is shrunk by a sample-size-aware factor (0.282x here; reaching full weight requires n≥60 months), since it remains the only estimator that captures nonlinear driver interactions without those interactions being hand-specified as products. Unshrunk XGBoost importances are retained in learned_driver_weights.csv.

Interval calibration

Prediction intervals are also checked, not just reported: Section 7 retrospectively rebuilds the bootstrap-ensemble interval at each backtest fold and measures how often the realized value actually fell inside it, against the interval's nominal coverage. As with the backtest MAPE, this calibration check runs on a small number of folds and should be read directionally.

Limitations

This model is a directional planning tool, trained on 22 months of history, not a substitute for a licensed rate benchmarking subscription or actual carrier quotes. Only 2 of 60 destination base rates are anchored to a real published tariff; the rest remain calibrated planning estimates. With this little history, deep sequence models (LSTM, Temporal Fusion Transformer, N-BEATS) were evaluated and rejected in favor of the Ridge/ElasticNet/Bayesian/XGBoost hybrid — see ARCHITECTURE.md for the full comparison and roadmap. Treat all lane-level dollar figures as directional, not quotable, rates.

Report generated from the IOFI v2 reusable pipeline (pipeline_runner.py → report_data.py → charts_v2.py → build_report_v2.py). See README.md and ARCHITECTURE.md for the full model design and monthly refresh process.

Appendix: Full Lane-Level Forecast (Model Median Case, USD/FEU)

All 180 origin-destination lanes (3 origins × 60 destinations), 3-month-ahead model median forecast. 5th/95th percentile detail for every lane is in lane_forecasts_v2.csv alongside this report.

180/180 lanes below are AI-checked (Source = AI): cross-referenced against an LLM's estimate of realistic current market rates, replacing the model's calibrated planning estimate for that lane. Remaining lanes (Source = Model) still use the calibrated estimate -- run pipeline/ai_rate_lookup.py to extend AI coverage. AI figures are still directional planning estimates, not carrier quotes.

Origin	Destination	Region	Current	3-mo Forecast	% Change	Source
ICD Dhannad	Buenos Aires	Africa / South America (Cape)	$5,088	$4,732	-7.0%	AI
ICD Dhannad	Cape Town	Africa / South America (Cape)	$3,673	$3,416	-7.0%	AI
ICD Dhannad	Durban	Africa / South America (Cape)	$3,647	$3,392	-7.0%	AI
ICD Dhannad	Lagos	Africa / South America (Cape)	$4,237	$3,940	-7.0%	AI
ICD Dhannad	Santos	Africa / South America (Cape)	$4,404	$4,096	-7.0%	AI
ICD Dhannad	Tema	Africa / South America (Cape)	$4,489	$4,175	-7.0%	AI
ICD Tihi	Buenos Aires	Africa / South America (Cape)	$4,871	$4,530	-7.0%	AI
ICD Tihi	Cape Town	Africa / South America (Cape)	$3,455	$3,213	-7.0%	AI
ICD Tihi	Durban	Africa / South America (Cape)	$3,429	$3,189	-7.0%	AI
ICD Tihi	Lagos	Africa / South America (Cape)	$4,019	$3,738	-7.0%	AI
ICD Tihi	Santos	Africa / South America (Cape)	$4,186	$3,893	-7.0%	AI
ICD Tihi	Tema	Africa / South America (Cape)	$4,272	$3,973	-7.0%	AI
Nhava Sheva (JNPT)	Buenos Aires	Africa / South America (Cape)	$4,602	$4,280	-7.0%	AI
Nhava Sheva (JNPT)	Cape Town	Africa / South America (Cape)	$3,187	$2,964	-7.0%	AI
Nhava Sheva (JNPT)	Durban	Africa / South America (Cape)	$3,161	$2,940	-7.0%	AI
Nhava Sheva (JNPT)	Lagos	Africa / South America (Cape)	$3,751	$3,488	-7.0%	AI
Nhava Sheva (JNPT)	Santos	Africa / South America (Cape)	$3,918	$3,644	-7.0%	AI
Nhava Sheva (JNPT)	Tema	Africa / South America (Cape)	$4,003	$3,723	-7.0%	AI
ICD Dhannad	Ambarli (Istanbul)	Europe (Mediterranean)	$4,343	$3,996	-8.0%	AI
ICD Dhannad	Barcelona	Europe (Mediterranean)	$4,360	$4,011	-8.0%	AI
ICD Dhannad	Genoa	Europe (Mediterranean)	$4,320	$3,974	-8.0%	AI
ICD Dhannad	Gioia Tauro	Europe (Mediterranean)	$4,624	$4,254	-8.0%	AI
ICD Dhannad	Marseille	Europe (Mediterranean)	$4,552	$4,188	-8.0%	AI
ICD Dhannad	Piraeus	Europe (Mediterranean)	$4,101	$3,773	-8.0%	AI
ICD Dhannad	Valencia	Europe (Mediterranean)	$4,396	$4,044	-8.0%	AI
ICD Tihi	Ambarli (Istanbul)	Europe (Mediterranean)	$4,111	$3,782	-8.0%	AI
ICD Tihi	Barcelona	Europe (Mediterranean)	$4,128	$3,798	-8.0%	AI
ICD Tihi	Genoa	Europe (Mediterranean)	$4,087	$3,760	-8.0%	AI
ICD Tihi	Gioia Tauro	Europe (Mediterranean)	$4,391	$4,040	-8.0%	AI
ICD Tihi	Marseille	Europe (Mediterranean)	$4,320	$3,974	-8.0%	AI
ICD Tihi	Piraeus	Europe (Mediterranean)	$3,869	$3,559	-8.0%	AI
ICD Tihi	Valencia	Europe (Mediterranean)	$4,163	$3,830	-8.0%	AI

Origin	Destination	Region	Current	3-mo Forecast	% Change	Source
Nhava Sheva (JNPT)	Ambarli (Istanbul)	Europe (Mediterranean)	$3,824	$3,518	-8.0%	AI
Nhava Sheva (JNPT)	Barcelona	Europe (Mediterranean)	$3,841	$3,534	-8.0%	AI
Nhava Sheva (JNPT)	Genoa	Europe (Mediterranean)	$3,801	$3,497	-8.0%	AI
Nhava Sheva (JNPT)	Gioia Tauro	Europe (Mediterranean)	$4,104	$3,776	-8.0%	AI
Nhava Sheva (JNPT)	Marseille	Europe (Mediterranean)	$4,033	$3,710	-8.0%	AI
Nhava Sheva (JNPT)	Piraeus	Europe (Mediterranean)	$3,582	$3,295	-8.0%	AI
Nhava Sheva (JNPT)	Valencia	Europe (Mediterranean)	$3,877	$3,567	-8.0%	AI
ICD Dhannad	Antwerp	Europe (North)	$5,130	$4,720	-8.0%	AI
ICD Dhannad	Felixstowe	Europe (North)	$5,281	$4,859	-8.0%	AI
ICD Dhannad	Hamburg	Europe (North)	$5,782	$5,319	-8.0%	AI
ICD Dhannad	Le Havre	Europe (North)	$5,473	$5,035	-8.0%	AI
ICD Dhannad	Rotterdam	Europe (North)	$5,355	$4,927	-8.0%	AI
ICD Tihi	Antwerp	Europe (North)	$4,898	$4,506	-8.0%	AI
ICD Tihi	Felixstowe	Europe (North)	$5,048	$4,644	-8.0%	AI
ICD Tihi	Hamburg	Europe (North)	$5,549	$5,105	-8.0%	AI
ICD Tihi	Le Havre	Europe (North)	$5,241	$4,822	-8.0%	AI
ICD Tihi	Rotterdam	Europe (North)	$5,123	$4,713	-8.0%	AI
Nhava Sheva (JNPT)	Antwerp	Europe (North)	$4,611	$4,242	-8.0%	AI
Nhava Sheva (JNPT)	Felixstowe	Europe (North)	$4,762	$4,381	-8.0%	AI
Nhava Sheva (JNPT)	Hamburg	Europe (North)	$5,262	$4,841	-8.0%	AI
Nhava Sheva (JNPT)	Le Havre	Europe (North)	$4,954	$4,558	-8.0%	AI
Nhava Sheva (JNPT)	Rotterdam	Europe (North)	$4,836	$4,449	-8.0%	AI
ICD Dhannad	Colombo	Indian Ocean Rim	$962	$924	-4.0%	AI
ICD Dhannad	Dar es Salaam	Indian Ocean Rim	$1,247	$1,197	-4.0%	AI
ICD Dhannad	Karachi	Indian Ocean Rim	$1,021	$980	-4.0%	AI
ICD Dhannad	Mombasa	Indian Ocean Rim	$1,197	$1,149	-4.0%	AI
ICD Dhannad	Port Louis	Indian Ocean Rim	$1,433	$1,376	-4.0%	AI
ICD Tihi	Colombo	Indian Ocean Rim	$800	$768	-4.0%	AI
ICD Tihi	Dar es Salaam	Indian Ocean Rim	$1,085	$1,042	-4.0%	AI
ICD Tihi	Karachi	Indian Ocean Rim	$859	$825	-4.0%	AI
ICD Tihi	Mombasa	Indian Ocean Rim	$1,035	$994	-4.0%	AI
ICD Tihi	Port Louis	Indian Ocean Rim	$1,271	$1,220	-4.0%	AI
Nhava Sheva (JNPT)	Colombo	Indian Ocean Rim	$600	$576	-4.0%	AI
Nhava Sheva (JNPT)	Dar es Salaam	Indian Ocean Rim	$886	$851	-4.0%	AI
Nhava Sheva (JNPT)	Karachi	Indian Ocean Rim	$659	$633	-3.9%	AI
Nhava Sheva (JNPT)	Mombasa	Indian Ocean Rim	$835	$802	-4.0%	AI
Nhava Sheva (JNPT)	Port Louis	Indian Ocean Rim	$1,071	$1,028	-4.0%	AI
ICD Dhannad	Busan	Intra-Asia	$1,399	$1,343	-4.0%	AI
ICD Dhannad	Ho Chi Minh City (Cat Lai)	Intra-Asia	$1,031	$990	-4.0%	AI

Origin	Destination	Region	Current	3-mo Forecast	% Change	Source
ICD Dhannad	Laem Chabang	Intra-Asia	$996	$956	-4.0%	AI
ICD Dhannad	Ningbo	Intra-Asia	$1,262	$1,212	-4.0%	AI
ICD Dhannad	Port Klang	Intra-Asia	$860	$826	-4.0%	AI
ICD Dhannad	Shanghai	Intra-Asia	$1,164	$1,117	-4.0%	AI
ICD Dhannad	Singapore	Intra-Asia	$897	$861	-4.0%	AI
ICD Dhannad	Yantian (Shenzhen)	Intra-Asia	$1,179	$1,132	-4.0%	AI
ICD Dhannad	Yokohama	Intra-Asia	$1,414	$1,357	-4.0%	AI
ICD Tihi	Busan	Intra-Asia	$1,234	$1,185	-4.0%	AI
ICD Tihi	Ho Chi Minh City (Cat Lai)	Intra-Asia	$866	$831	-4.0%	AI
ICD Tihi	Laem Chabang	Intra-Asia	$831	$798	-4.0%	AI
ICD Tihi	Ningbo	Intra-Asia	$1,097	$1,053	-4.0%	AI
ICD Tihi	Port Klang	Intra-Asia	$695	$667	-4.0%	AI
ICD Tihi	Shanghai	Intra-Asia	$999	$959	-4.0%	AI
ICD Tihi	Singapore	Intra-Asia	$732	$703	-4.0%	AI
ICD Tihi	Yantian (Shenzhen)	Intra-Asia	$1,014	$973	-4.0%	AI
ICD Tihi	Yokohama	Intra-Asia	$1,248	$1,198	-4.0%	AI
Nhava Sheva (JNPT)	Busan	Intra-Asia	$1,031	$990	-4.0%	AI
Nhava Sheva (JNPT)	Ho Chi Minh City (Cat Lai)	Intra-Asia	$662	$636	-3.9%	AI
Nhava Sheva (JNPT)	Laem Chabang	Intra-Asia	$627	$602	-4.0%	AI
Nhava Sheva (JNPT)	Ningbo	Intra-Asia	$893	$857	-4.0%	AI
Nhava Sheva (JNPT)	Port Klang	Intra-Asia	$492	$472	-4.1%	AI
Nhava Sheva (JNPT)	Shanghai	Intra-Asia	$795	$763	-4.0%	AI
Nhava Sheva (JNPT)	Singapore	Intra-Asia	$528	$507	-4.0%	AI
Nhava Sheva (JNPT)	Yantian (Shenzhen)	Intra-Asia	$811	$779	-3.9%	AI
Nhava Sheva (JNPT)	Yokohama	Intra-Asia	$1,045	$1,003	-4.0%	AI
ICD Dhannad	Bandar Abbas	Middle East	$1,431	$1,359	-5.0%	AI
ICD Dhannad	Dammam	Middle East	$1,409	$1,339	-5.0%	AI
ICD Dhannad	Hamad Port (Doha)	Middle East	$1,295	$1,230	-5.0%	AI
ICD Dhannad	Jebel Ali	Middle East	$1,323	$1,257	-5.0%	AI
ICD Dhannad	Kuwait City	Middle East	$1,317	$1,251	-5.0%	AI
ICD Dhannad	Sohar	Middle East	$1,341	$1,274	-5.0%	AI
ICD Tihi	Bandar Abbas	Middle East	$1,266	$1,203	-5.0%	AI
ICD Tihi	Dammam	Middle East	$1,244	$1,182	-5.0%	AI
ICD Tihi	Hamad Port (Doha)	Middle East	$1,130	$1,074	-5.0%	AI
ICD Tihi	Jebel Ali	Middle East	$1,158	$1,100	-5.0%	AI
ICD Tihi	Kuwait City	Middle East	$1,152	$1,094	-5.0%	AI
ICD Tihi	Sohar	Middle East	$1,176	$1,117	-5.0%	AI
Nhava Sheva (JNPT)	Bandar Abbas	Middle East	$1,063	$1,010	-5.0%	AI
Nhava Sheva (JNPT)	Dammam	Middle East	$1,041	$989	-5.0%	AI

Origin	Destination	Region	Current	3-mo Forecast	% Change	Source
Nhava Sheva (JNPT)	Hamad Port (Doha)	Middle East	$926	$880	-5.0%	AI
Nhava Sheva (JNPT)	Jebel Ali	Middle East	$955	$907	-5.0%	AI
Nhava Sheva (JNPT)	Kuwait City	Middle East	$948	$901	-5.0%	AI
Nhava Sheva (JNPT)	Sohar	Middle East	$972	$923	-5.0%	AI
ICD Dhannad	Auckland	Pacific (US West / Oceania)	$2,833	$2,691	-5.0%	AI
ICD Dhannad	Los Angeles/Long Beach	Pacific (US West / Oceania)	$7,131	$6,489	-9.0%	AI
ICD Dhannad	Melbourne	Pacific (US West / Oceania)	$2,716	$2,580	-5.0%	AI
ICD Dhannad	Oakland	Pacific (US West / Oceania)	$7,380	$6,716	-9.0%	AI
ICD Dhannad	Seattle/Tacoma	Pacific (US West / Oceania)	$6,864	$6,246	-9.0%	AI
ICD Dhannad	Sydney	Pacific (US West / Oceania)	$2,928	$2,782	-5.0%	AI
ICD Dhannad	Vancouver	Pacific (US West / Oceania)	$3,089	$2,935	-5.0%	AI
ICD Tihi	Auckland	Pacific (US West / Oceania)	$2,653	$2,520	-5.0%	AI
ICD Tihi	Los Angeles/Long Beach	Pacific (US West / Oceania)	$6,696	$6,093	-9.0%	AI
ICD Tihi	Melbourne	Pacific (US West / Oceania)	$2,536	$2,409	-5.0%	AI
ICD Tihi	Oakland	Pacific (US West / Oceania)	$6,946	$6,321	-9.0%	AI
ICD Tihi	Seattle/Tacoma	Pacific (US West / Oceania)	$6,429	$5,850	-9.0%	AI
ICD Tihi	Sydney	Pacific (US West / Oceania)	$2,748	$2,611	-5.0%	AI
ICD Tihi	Vancouver	Pacific (US West / Oceania)	$2,909	$2,764	-5.0%	AI
Nhava Sheva (JNPT)	Auckland	Pacific (US West / Oceania)	$2,431	$2,309	-5.0%	AI
Nhava Sheva (JNPT)	Los Angeles/Long Beach	Pacific (US West / Oceania)	$6,160	$5,606	-9.0%	AI
Nhava Sheva (JNPT)	Melbourne	Pacific (US West / Oceania)	$2,314	$2,198	-5.0%	AI
Nhava Sheva (JNPT)	Oakland	Pacific (US West / Oceania)	$6,409	$5,832	-9.0%	AI
Nhava Sheva (JNPT)	Seattle/Tacoma	Pacific (US West / Oceania)	$5,893	$5,363	-9.0%	AI
Nhava Sheva (JNPT)	Sydney	Pacific (US West / Oceania)	$2,526	$2,400	-5.0%	AI
Nhava Sheva (JNPT)	Vancouver	Pacific (US West / Oceania)	$2,687	$2,553	-5.0%	AI
ICD Dhannad	Aqaba	Red Sea / East Africa	$2,065	$1,941	-6.0%	AI
ICD Dhannad	Djibouti	Red Sea / East Africa	$2,105	$1,979	-6.0%	AI
ICD Dhannad	Jeddah	Red Sea / East Africa	$1,732	$1,628	-6.0%	AI
ICD Dhannad	Port Sudan	Red Sea / East Africa	$2,300	$2,162	-6.0%	AI
ICD Dhannad	Suez	Red Sea / East Africa	$2,282	$2,145	-6.0%	AI
ICD Tihi	Aqaba	Red Sea / East Africa	$1,878	$1,765	-6.0%	AI
ICD Tihi	Djibouti	Red Sea / East Africa	$1,918	$1,803	-6.0%	AI
ICD Tihi	Jeddah	Red Sea / East Africa	$1,545	$1,452	-6.0%	AI
ICD Tihi	Port Sudan	Red Sea / East Africa	$2,112	$1,985	-6.0%	AI
ICD Tihi	Suez	Red Sea / East Africa	$2,095	$1,969	-6.0%	AI
Nhava Sheva (JNPT)	Aqaba	Red Sea / East Africa	$1,646	$1,547	-6.0%	AI
Nhava Sheva (JNPT)	Djibouti	Red Sea / East Africa	$1,686	$1,585	-6.0%	AI
Nhava Sheva (JNPT)	Jeddah	Red Sea / East Africa	$1,314	$1,235	-6.0%	AI
Nhava Sheva (JNPT)	Port Sudan	Red Sea / East Africa	$1,881	$1,768	-6.0%	AI

Origin	Destination	Region	Current	3-mo Forecast	% Change	Source
Nhava Sheva (JNPT)	Suez	Red Sea / East Africa	$1,864	$1,752	-6.0%	AI
ICD Dhannad	Baltimore	US East Coast	$8,674	$7,807	-10.0%	AI
ICD Dhannad	Charleston	US East Coast	$8,008	$7,207	-10.0%	AI
ICD Dhannad	New York/New Jersey	US East Coast	$7,778	$7,000	-10.0%	AI
ICD Dhannad	Norfolk	US East Coast	$8,104	$7,294	-10.0%	AI
ICD Dhannad	Savannah	US East Coast	$7,657	$6,891	-10.0%	AI
ICD Tihi	Baltimore	US East Coast	$8,336	$7,502	-10.0%	AI
ICD Tihi	Charleston	US East Coast	$7,670	$6,903	-10.0%	AI
ICD Tihi	New York/New Jersey	US East Coast	$7,441	$6,697	-10.0%	AI
ICD Tihi	Norfolk	US East Coast	$7,767	$6,990	-10.0%	AI
ICD Tihi	Savannah	US East Coast	$7,319	$6,587	-10.0%	AI
Nhava Sheva (JNPT)	Baltimore	US East Coast	$7,920	$7,128	-10.0%	AI
Nhava Sheva (JNPT)	Charleston	US East Coast	$7,254	$6,529	-10.0%	AI
Nhava Sheva (JNPT)	New York/New Jersey	US East Coast	$7,024	$6,322	-10.0%	AI
Nhava Sheva (JNPT)	Norfolk	US East Coast	$7,351	$6,616	-10.0%	AI
Nhava Sheva (JNPT)	Savannah	US East Coast	$6,903	$6,213	-10.0%	AI
ICD Dhannad	Balboa	US Gulf / Caribbean / Latin America	$3,961	$3,723	-6.0%	AI
ICD Dhannad	Cartagena	US Gulf / Caribbean / Latin America	$4,231	$3,977	-6.0%	AI
ICD Dhannad	Houston	US Gulf / Caribbean / Latin America	$8,662	$7,796	-10.0%	AI
ICD Dhannad	Kingston	US Gulf / Caribbean / Latin America	$3,902	$3,668	-6.0%	AI
ICD Dhannad	New Orleans	US Gulf / Caribbean / Latin America	$8,420	$7,578	-10.0%	AI
ICD Tihi	Balboa	US Gulf / Caribbean / Latin America	$3,758	$3,533	-6.0%	AI
ICD Tihi	Cartagena	US Gulf / Caribbean / Latin America	$4,028	$3,786	-6.0%	AI
ICD Tihi	Houston	US Gulf / Caribbean / Latin America	$8,325	$7,492	-10.0%	AI
ICD Tihi	Kingston	US Gulf / Caribbean / Latin America	$3,699	$3,477	-6.0%	AI
ICD Tihi	New Orleans	US Gulf / Caribbean / Latin America	$8,082	$7,274	-10.0%	AI
Nhava Sheva (JNPT)	Balboa	US Gulf / Caribbean / Latin America	$3,509	$3,298	-6.0%	AI
Nhava Sheva (JNPT)	Cartagena	US Gulf / Caribbean / Latin America	$3,779	$3,552	-6.0%	AI
Nhava Sheva (JNPT)	Houston	US Gulf / Caribbean / Latin America	$7,909	$7,118	-10.0%	AI
Nhava Sheva (JNPT)	Kingston	US Gulf / Caribbean / Latin America	$3,449	$3,242	-6.0%	AI
Nhava Sheva (JNPT)	New Orleans	US Gulf / Caribbean / Latin America	$7,666	$6,899	-10.0%	AI